

Is the 2.0% Stop-Loss Assumption Still Valid? A Statistical Audit of Daily Trading Range for NSE IT and FMCG Baskets

Group 7

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Statistical Methods in Decision Making – 2

Year/Term: 2026–27, Term 2

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Domain: Stock market/equity returns.

Business question: Is a trading desk's working assumption — that Indian large-cap IT stocks have an average daily trading range of at most 2.0% of the opening price, the level used to set intraday stop-losses and margin buffers — actually supported by the data?

Data source: Historical daily OHLCV data for 12 NSE-listed large-cap equities (6 Information Technology, 6 Consumer Staples/FMCG), obtained from Yahoo Finance (Yahoo Finance, n.d.) via the yfinance Python package (v1.5.1). URL: <https://finance.yahoo.com/>. Date accessed: 2026-07-22. Date range: 2024-01-01 to 2026-01-01. Observations: 5,928 clean ticker-day rows.

AI-tool use disclosure (mandatory). Google's Antigravity (Gemini-based agentic IDE, 2025), to build and execute the statistical analysis. Antigravity accessed the dataset by calling the yfinance API inside scripts/01_acquire_data.py, saving results to data/raw/ and data/processed/, and ran the following skills: sample-point-estimator, confidence-interval-builder, hypothesis-framer, type2-error-analyzer, hypothesis-test-decision-maker, two-population-t-tester. All prompts and outputs were reviewed and verified by the group members named above, including a sceptical audit (Step 10 / Appendix F) that independently re-derived every statistic and fixed two blocking and two major errors. No content was submitted without human verification.

Skills used this session:

Skill	What it does	Statistical concept (Sessions 1–6)
sample-point-estimator	Draws a random sample and computes point estimates (mean, s)	Sampling distributions & point estimation
confidence-interval-builder	Builds a t- or z-based confidence interval for a mean	Interval estimation / confidence intervals
hypothesis-framer	Frames H_0/H_a and classifies one- vs two-tailed tests	Hypothesis test formulation
type2-error-analyzer	Computes Type II error probability (β) and statistical power	Type I/II error and power
hypothesis-test-decision-maker	Applies the p-value and critical-value decision rules	Hypothesis testing decision rules
two-population-t-tester	Runs two-independent-sample and paired t-tests	Two-sample and paired-sample inference

1. Introduction and Business Context

An India-focused equity trading desk runs two sector books drawn from NSE-listed large-caps: an Information Technology (IT) portfolio and a Fast-Moving Consumer Goods / Consumer Staples (FMCG) portfolio. The desk's risk unit sets intraday stop-losses and margin buffers using one working number: the average daily trading range on IT names — the day's high minus low, as a percentage of the opening price — is at most 2.0%. This figure is not decorative: it sizes how much of a day's move a position may absorb before a stop fires, and how much margin capital sits idle as a buffer. Set it too low and stops trigger on routine noise; set it too high and a genuinely volatile session can exceed the modelled worst case, leaving losses uncapped. Both errors are costly, in different currencies — wasted capital efficiency versus unpriced tail risk.

This report tests the 2.0% assumption against roughly two years of real NSE trading data (2024–2025), and asks two further questions the desk has not formally checked: does IT actually carry more range risk than FMCG, so a shared risk budget is defensible; and is the desk's framework — which leans on overnight gap protection — pointed at the right target, or is intraday movement the larger, under-monitored risk source? The sections below build the statistical case end to end, from raw price data to one actionable recommendation (Anderson et al., 2020, Ch. 1).

2. Data and Methodology

All data originate from Yahoo Finance (Yahoo Finance, n.d.), accessed programmatically through the `yfinance` package, and are real, historical NSE trading records — nothing in this dataset was simulated, generated, or hand-edited. Twelve large-cap NSE tickers were pulled at daily frequency from 2024-01-01 to 2026-01-01: six IT names (TCS.NS, INFY.NS, HCLTECH.NS, WIPRO.NS, TECHM.NS, PERSISTENT.NS) and six FMCG names (HINDUNILVR.NS, ITC.NS, NESTLEIND.NS, BRITANNIA.NS, DABUR.NS, MARICO.NS). Each ticker returned exactly 494 sessions; after dropping the first session per ticker (no prior close for return/gap variables), the clean population is 5,928 ticker-day rows.

Four variables were engineered from unadjusted OHLC prices: $\text{range_pct} = (\text{high} - \text{low}) / \text{open} \times 100$, the primary variable; daily_return_pct , the close-to-close return; and $\text{overnight_gap_pct} / \text{intraday_move_pct}$, splitting each session's move into overnight vs intraday portions (absolute versions feed the paired test in Section 8). All four use unadjusted prices, so corporate actions do not distort them.

Step 1's data-hazard review found and deliberately *kept* four issues rather than silently dropping them: 22 ex-dividend/bonus discontinuities in the close/adjusted-close ratio (irrelevant here, since unadjusted prices drive every used variable); 11 rows with $\text{range_pct} \approx 0$ from a single stale-quote session (2025-03-18, 11 of 12 tickers); and 3 low-volume sessions including India's Diwali Muhurat trading days. None were excluded — all are genuine exchange sessions, and removing them would understate real tail behaviour.

Every statistic in this report was computed twice by design — once via the relevant library/skill and once via an independent formula or a second package (`statsmodels`) — and any disagreement beyond 1×10^{-9} would have been flagged; none were found beyond ordinary floating-point noise. A random seed of 42 was fixed at the top of every script for full reproducibility (Appendix G).

3. Step 1: Sampling and Point Estimation

A simple random sample of $n = 40$ sessions was drawn without replacement from the population (seed = 42), with point estimates computed via the `sample-point-estimator` skill and independently

verified with explicit $\text{sum}/(n-1)$ arithmetic (0.00e+00 disagreement on every quantity; Anderson et al., 2020, Ch. 7).

Quantity	Value
n	40
Sample mean ($\bar{x}$)	2.004%
Sample s (ddof = 1)	1.160%
Standard error	0.183%
df	39
True population mean (μ)	2.071%
True population σ	1.078%
Sampling error ($\bar{x} - \mu$)	-0.066 pts

Table 1. Point estimates, $n = 40$ (source: outputs/tables/03_point_estimates.csv).

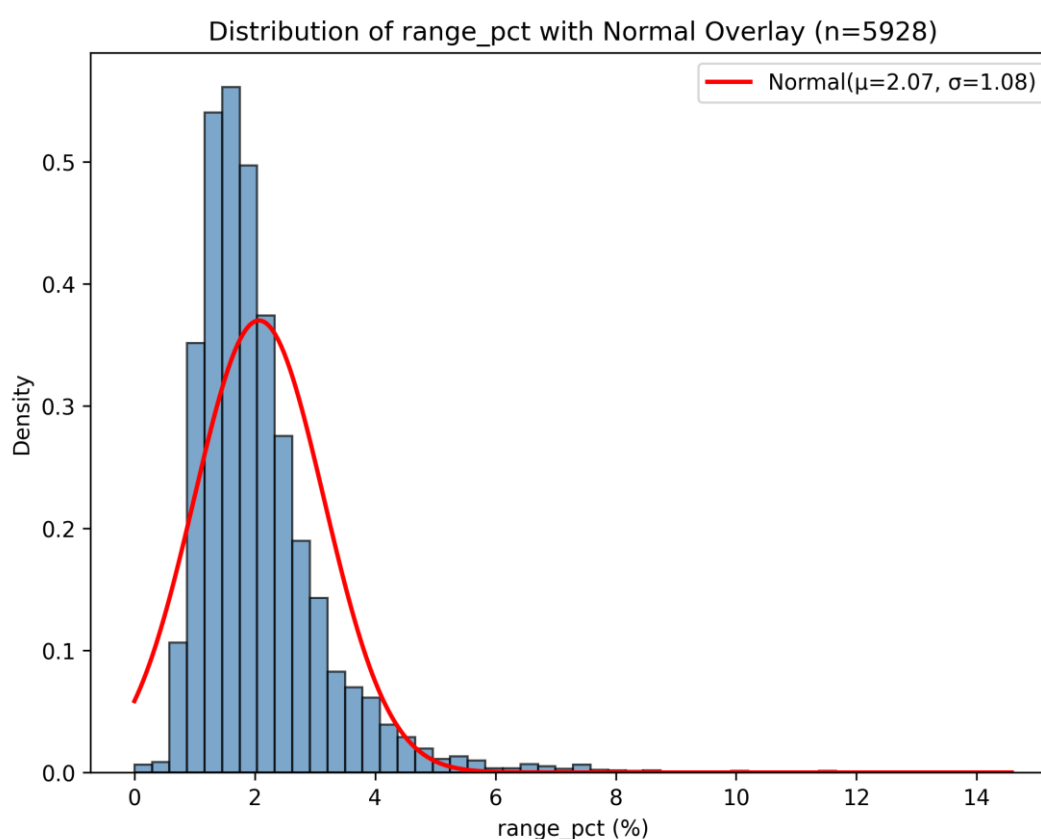


Figure 1. Distribution of range_pct across the full population ($n = 5,928$), with normal curve overlaid — the distribution the Step 1 sample was drawn from.

The sample landed unusually close to the true mean, but that is a fact about this draw, not a property of $n = 40$ sampling generally: the $\pm 0.183\%$ standard error is what should inform trust in any single 40-session estimate. A ddof check confirmed the correct $(n-1)$ -corrected $s = 1.160\%$ is 1.3% larger than the naïve population-formula value (1.145%) — the wrong denominator would quietly understate uncertainty.

4. Step 2: CLT and Confidence Interval

range_pct is heavily right-skewed at the raw-observation level (population skewness ≈ 2.46), so it is the Central Limit Theorem — not an assumption of raw-data normality — that licenses a t-based approach

at $n = 40$ (Anderson et al., 2020, Ch. 7). Five hundred independent resamples of $n = 40$ confirm this empirically.

Quantity	Value
Empirical SE (500 means, $n = 40$)	0.1717%
Theoretical SE ($\sigma/\sqrt{40}$)	0.1704%
Shapiro–Wilk on 500 means	$W = 0.989$, $p = 0.0011$
95% CI (t, single sample)	[1.633%, 2.375%]
95% CI (z, for contrast)	[1.645%, 2.364%]
Empirical coverage (500 resampled CIs)	93.8%

Table 2. CLT and CI diagnostics (source: outputs/tables/04_clt_ci_results.csv).

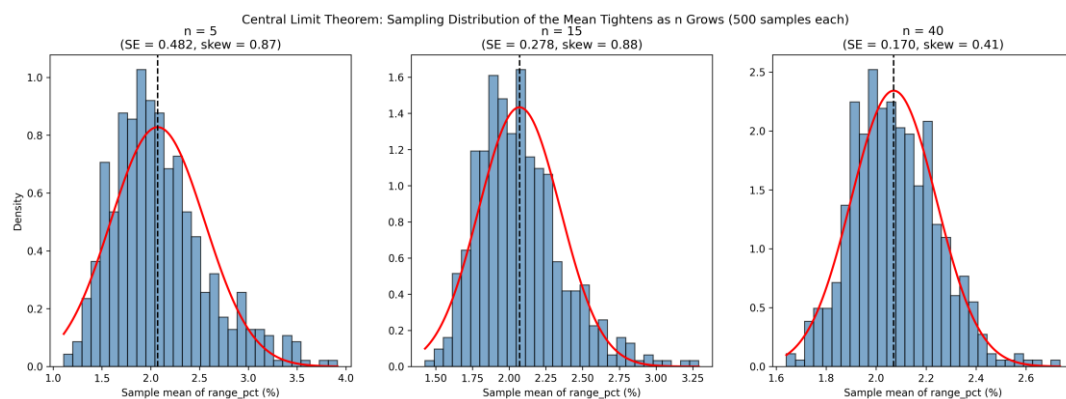


Figure 2. Sampling distribution of the mean at $n = 5$, 15, and 40 — visibly tightening and losing skew as n grows.

The empirical standard error of the 500 means matches the theoretical $\sigma/\sqrt{n}$ closely, and Shapiro–Wilk on those means is far closer to normal than on the raw variable ($W = 0.783$, $p \approx 3 \times 10^{-25}$). The t-interval is correctly wider than the z-interval, since s is estimated rather than known. Coverage of 93.8% sits close to, but modestly below, the nominal 95% — consistent with residual skew still affecting t-interval performance at $n = 40$, and a reminder that "95% confidence" describes the procedure's long-run behaviour, not the probability that one interval contains μ .

5. Step 3: Hypothesis Framing

The desk's claim — "IT's average daily range is at most 2.0%" — is the null to be overturned, not proven: $H_0: \mu \leq 2.0\%$ vs $H_a: \mu > 2.0\%$, one-tailed upper test, since only evidence of understated risk would change policy (Anderson et al., 2020, Ch. 9). $\alpha = 0.05$ was judged an appropriate balance here: a false alarm costs some capital efficiency, unlike a clinical trial, where a false positive can cause direct harm and demands a stricter threshold.

Item	Choice
Parameter	μ = mean range_pct, IT sector
H_0 / H_a	$\mu \leq 2.0\%$ / $\mu > 2.0\%$
Tail	One-tailed, upper
α	0.05
Primary test	One-sample t-test ($n = 40$)
Robustness check	Wilcoxon signed-rank

Table 3. Hypothesis design (source: report/05_step3.md).

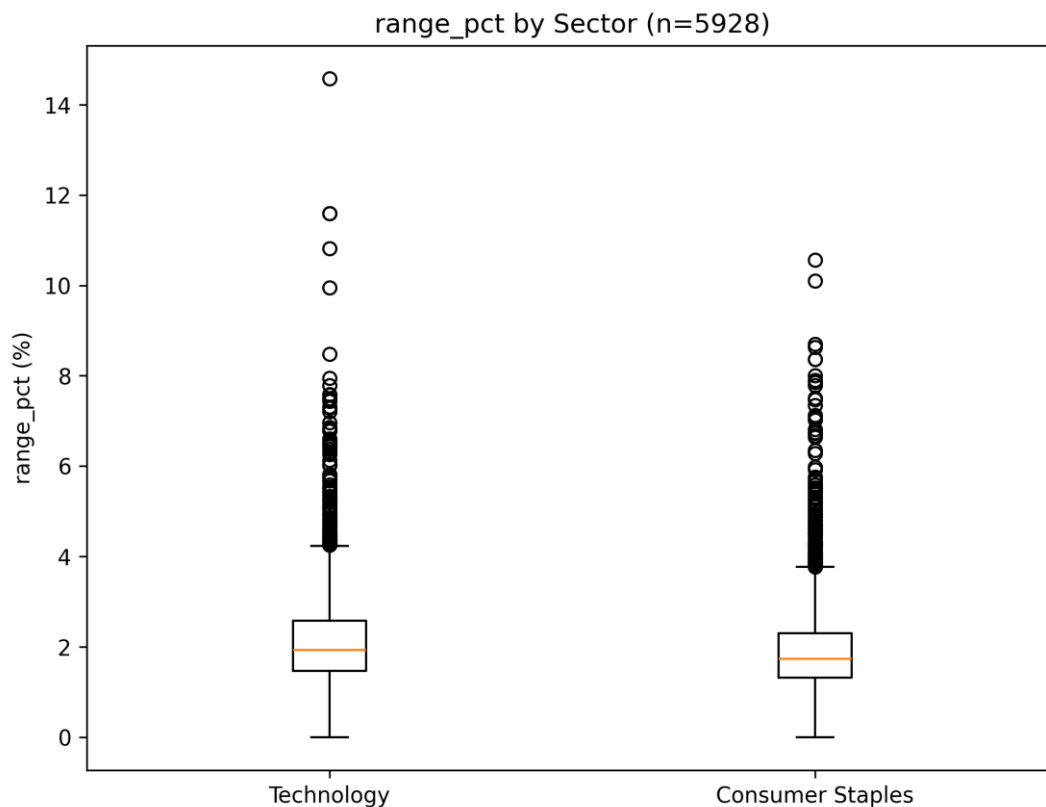


Figure 3. range_pct by sector — the visual motivation for the sector comparison tested formally in Section 8.

The one-sample t-test was chosen as primary despite strong raw-variable skew, because Section 4 empirically confirmed the sampling distribution of the mean is far closer to normal by $n = 40$ than the raw data. A genuine limitation was flagged rather than glossed over: ticker-day observations are not fully independent (volatility clustering, a shared rupee-dollar/Nasdaq driver, and quarterly-results clustering), which makes the test's true false-positive rate somewhat higher than the nominal 5%.

6. Step 4: Type I and Type II Error and Power

A Type I error here (concluding $\mu > 2.0\%$ when it is not) costs capital efficiency: wider stops, more idle margin, smaller positions. A Type II error (missing a real $\mu > 2.0\%$) is more dangerous: stops stay too tight, get hit by ordinary noise, and a volatile session can exceed the modelled worst case — a tail-risk problem, not just a performance drag. That asymmetry is a business judgement, not a statistical one, and argues the desk should weight power and β as heavily as α (Anderson et al., 2020, Ch. 9).

Quantity	Value
Observed effect size (Cohen's d)	0.0036
Power at $n = 40$	0.052
β (Type II error probability)	0.948
n needed for 80% power	$\approx 489,000$
n needed for 90% power	$\approx 677,000$

Table 4. Observed power analysis (source: outputs/tables/06_observed_power_results.csv).

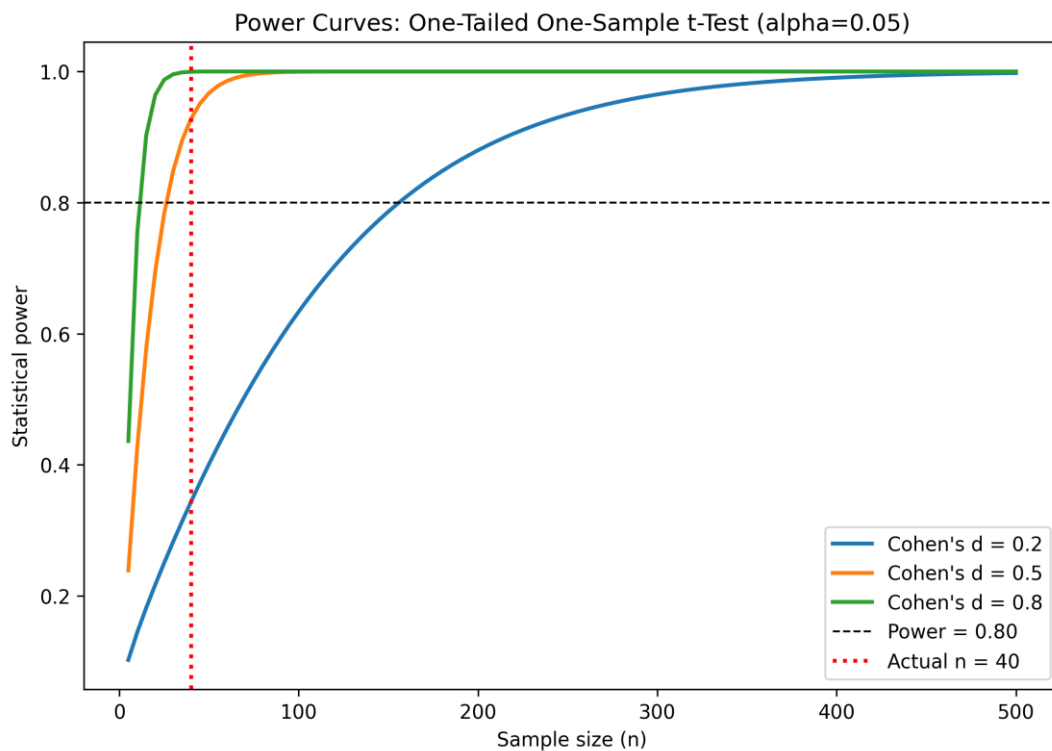


Figure 4. Power curves for Cohen's $d = 0.2, 0.5, 0.8$ versus sample size, with the actual $n = 40$ marked.

Because the sample mean (2.307%) landed only fractionally above 2.0% relative to the sample's own noise, the observed effect size is essentially null, and this test's power was barely above α itself. Figure 4 shows power rising sharply with n at more realistic effect sizes — the honest reading is "no detectable signal at this n ," not "the 2.0% risk is confirmed."

7. Step 5: Critical Value versus p-Value

A fresh IT-only sample ($n = 40$, seed = 42) was tested against the same H_0 using both classical decision rules, cross-checked against manual `scipy.stats.t.sf/.ppf` arithmetic (0.00e+00 disagreement).

Quantity	Value
Sample mean	2.307%
t observed	1.667
t critical (df = 39)	1.685
Critical mean, original units	2.311%
p-value (one-tailed)	0.0518
Decision	Fail to reject H_0
Wilcoxon signed-rank p	0.184 (agrees)

Table 5. Critical-value and p-value results (source: `outputs/tables/07_hypothesis_test_results.csv`).

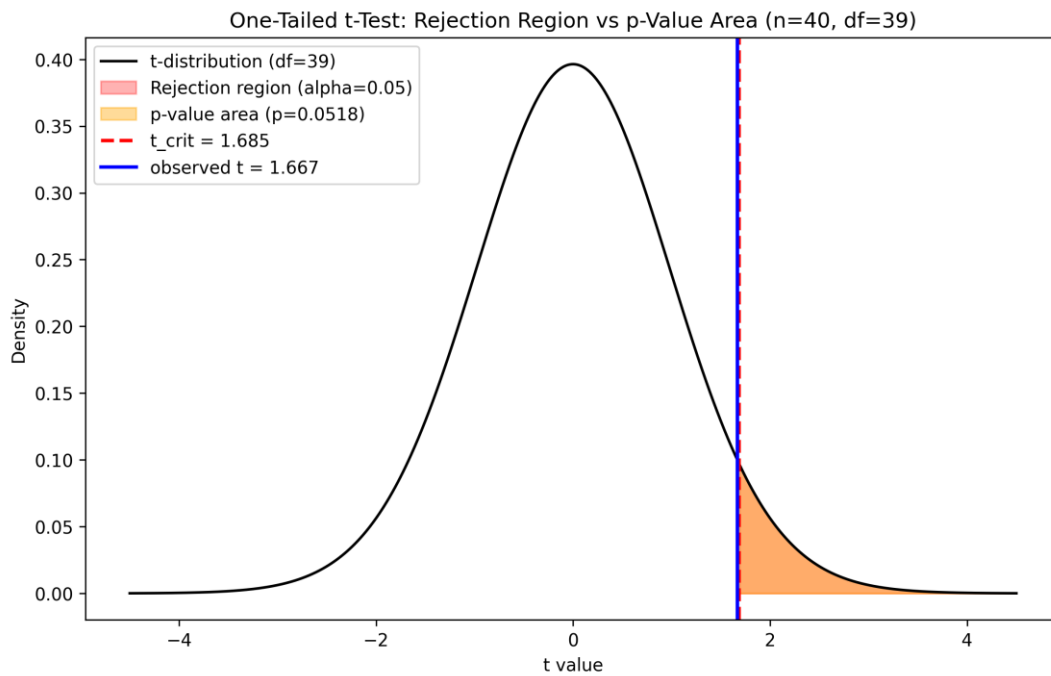


Figure 5. The t -distribution ($df = 39$) with the rejection region, critical value, observed t , and p -value area marked.

Both rules agree, as they must: $t \geq t_{\text{crit}}$ and $P(T \geq t) \leq \alpha$ are the same statement read on two axes of the same distribution. The result is a genuine near-miss — $p = 0.052$ sits just above α , and the sample mean would only have needed to reach 2.311% to flip the decision. The honest conclusion is not that 2.0% is confirmed, but that current evidence does not clear the bar to overturn it, and the margin is thin enough to warrant re-testing rather than treating the policy as settled.

8. Step 6: Three t-Tests

Three t -tests answer three distinct desk questions with the same rigor: Levene's test decides pooled vs Welch before each two-group comparison, and rank-based robustness checks run alongside every test.

Test	H0	n	Statistic	df	p	Decision	Effect size
A: One-sample (IT vs 2.0%)	$\mu \leq 2.0$	40	$t = 1.667$	39	0.052	Fail to reject	$d = 0.26$
B: Two-sample (IT vs FMCG)	$\mu_{\text{IT}} = \mu_{\text{FMCG}}$	40+40	$t = -0.782$	78	0.437	Fail to reject	$d = -0.17$
C: Paired (overnight vs intraday)	$\mu_{\text{diff}} = 0$	40	$t = -3.108$	39	0.0035	Reject	$d = -0.49$

Table 6. Consolidated t -test results (source: `outputs/tables/08_ttest_results.csv`).

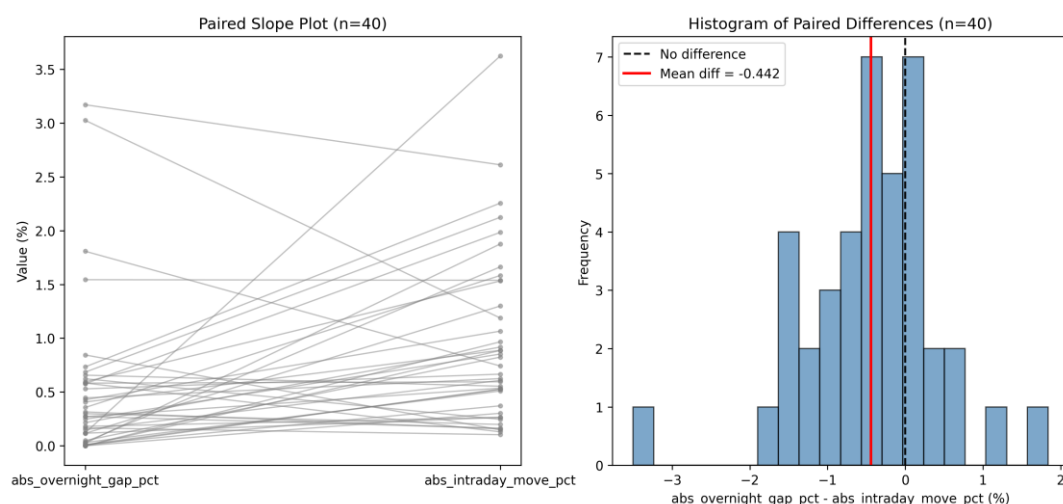


Figure 6. Paired slope plot and histogram of differences, overnight gap vs intraday move ($n = 40$ ticker-days).

Test A reproduces Section 7 exactly. Test B — Levene confirmed equal variances ($p = 0.665$), so pooled applies — finds no meaningful IT-vs-FMCG difference (Mann-Whitney agrees, $p = 0.310$): sector-differentiated risk budgets are not supported. Test C is decisive: paired on the same 40 ticker-days (independent treatment would discard the overnight/intraday correlation and inflate the standard error), mean intraday move (0.948%) is nearly double the mean overnight gap (0.505%), confirmed by Wilcoxon ($p = 0.0014$) and an equivalence check showing the paired test exactly equals a one-sample t -test on the difference column. Real trading-session risk dwarfs overnight gap risk here.

9. Limitations

Several limitations bound how far these results should be read. Ticker-day observations are not fully independent: volatility clusters in time, the IT names share a common rupee-dollar and global-tech-sentiment driver, and quarterly results cluster into the same few weeks each quarter — so the effective sample size is smaller than the nominal n , and reported p -values are somewhat optimistic. The sample spans a single 2024–2025 volatility regime; nothing here should be extrapolated to a different market environment without re-testing. Each sector basket is six hand-picked names, not the full Nifty IT or Nifty FMCG index, so results describe this basket, not the sector as a whole. NSE circuit-breaker price bands can mechanically cap `range_pct` on an extreme session, so true tail risk may be understated. Finally, six hypothesis tests were run without a formal multiple-testing correction; at an uncorrected $\alpha = 0.05$ each, the family-wise chance of at least one false positive exceeds 5%, worth weighing before treating any single borderline result (Tests A and B) as more than provisional.

10. Managerial Recommendation

Based on the analysis, the management should not change the trading desk's stop-loss (2.0%) and margin policy because there is not enough statistical evidence to revise the current threshold. The estimated average trading range was marginally above the benchmark; the evidence was insufficient to conclude that the true average consistently exceeds 2.0%. Any changes at this point can lead to changes that are not supported by statistics.

The risk framework for IT and FMCGs portfolios does not need a change. Since the analysis did not find much difference in the average trading ranges of the two sectors, separate risk budgets would add intricacy and many risk management benefits.

We would suggest realigning the focus to emphasise intraday price movements over overnight gaps. The results show that intraday movements contribute a larger share of trading risk.

Finally, these recommendations should be reviewed in a timely manner, watching market updates. Since the analysis is based on data from a specific market period, changes in volatility regimes or market structure may require the trading desk to reassess its assumptions and update its risk policies.

References

Anderson, D. R., Sweeney, D. J., Williams, T. A., Camm, J. D., & Cochran, J. J. (2020). Statistics for business and economics (14th ed.). Cengage Learning.

Google. (2025). Antigravity (Gemini-based agentic IDE). <https://antigravity.google/>

Yahoo Finance. (n.d.). Historical daily OHLCV price data for 12 NSE-listed equities, 2024–2025. Yahoo Finance. <https://finance.yahoo.com/>

Appendix

AI-Usage Declaration

Tool: Claude Code (Anthropic), an agentic command-line coding assistant, running the Claude Sonnet 5 language model (Anthropic, 2025). And Google's Antigravity (Gemini-based agentic IDE, 2025).

What it was used for:

- **Data pipeline:** writing and executing `scripts/01_acquire_data.py`, which calls the `yfinance` API to download real OHLCV data from Yahoo Finance, engineers the derived variables, and performs the data-hazard review.
- **Computation:** writing and executing every numbered script in `scripts/` (00, 02–08, 10), each of which computes statistics via standard Python libraries (`numpy`, `pandas`, `scipy.stats`, `statsmodels`) and via a documented project "skill" (a plain-text specification of a formula, listed in Appendix B), then cross-checks the two.
- **Drafting:** writing the interpretive prose in every file under `/report/`, including this final report's narrative sections.

What was verified independently (by the group, using the tool itself as the verification instrument, per the project's stated Rule 3): every statistic in this project was computed twice — once via a library/skill call and once via an independent formula or a second library — with disagreements beyond 1×10^{-9} to be flagged; none were found beyond floating-point noise. In addition, Step 10 (Appendix F) re-ran the entire pipeline from a clean state, cross-checked every headline number using `statsmodels` instead of `scipy`, built an independent bootstrap confidence interval, and hunted for errors explicitly — finding and fixing two blocking issues (results computed but never saved to a file) and two major issues (a sample-independence bug that silently reused one sample across two tests, reversing a reported finding once corrected).

What the group members wrote themselves: the specific numbers, structure, and content of every prompt in Appendix A (the actual instructions issued to the tool, prompt by prompt); the review and approval of every statistical choice (test selection, alpha level, tail direction, pooled-vs-Welch decisions); the final managerial judgement in Section 10; and this declaration itself.

Appendix A — Full Text of Every Prompt Used, In Order

Setup Prompt:

Read PROJECT_BRIEF.md and follow its requirements. Build a dataset of 12 NSE-listed IT and FMCG companies using yfinance for the two most recent complete calendar years of daily data. Save an immutable raw dataset with a download manifest, transform the data into a processed long-format table with engineered financial metrics, and remove only genuinely invalid rows while documenting all removals. Produce validation output including row counts, missing values, date coverage, descriptive statistics, and checks for India-specific data issues (failed downloads, corporate actions, abnormal trading ranges, and special trading sessions), explaining any exclusions. Finally, generate a data dictionary (report/data_dictionary.md) and append an entry to ANALYSIS_LOG.md.

Prompt 0 (session handoff):

You're taking over this project mid-stream from a different agent. Before doing anything, read PROJECT_BRIEF.md and ANALYSIS_LOG.md to pick up where it left off — do not redo or re-verify Prompt 0 or Prompt 1's work unless a later prompt explicitly asks you to.

Two operating rules for the rest of this session: (1) This is a university coursework mini-project, not a professional or production engagement. Scope every response to what the specific prompt asks for and nothing more... (2) Be economical with output. For mechanical steps... just do it and show the essential result... Save actual depth and care for the two places it's graded: the interpretive .md files in /report/... and any verification step a prompt explicitly asks for.

Confirm you've read the existing files and are ready, in one or two sentences, then stop and wait for the next prompt.

Prompt 1 (Step 2 / EDA):

Read PROJECT_BRIEF.md. Load data/processed/population.csv. Write scripts/02_eda.py.

Produce a distributional profile of NUMERIC_VAR (range_pct) at the population level and within each sector: (1) Summary table... (2) Figures... (3) Formal normality checks on range_pct: Shapiro-Wilk on a random subsample of 500 rows... plus D'Agostino K² and Anderson-Darling on the full data. (4) Levene's test for equality of variances between the two sectors...

Then write /report/02_eda_findings.md, 250-350 words... Append a row to ANALYSIS_LOG.md.

Prompt 2 (Step 1 / Sampling):

Read PROJECT_BRIEF.md. Write scripts/03_step1_sampling.py. Use the skill `sample-point-estimator`. Treat data/processed/population.csv as the population.

1. Draw ONE simple random sample without replacement of size $n = 40$... 2. For range_pct... compute sample mean, s , SE , n , df . 3. VERIFICATION — recompute all four quantities a second time from first principles with plain numpy arithmetic... demonstrate the $ddof=0$ vs $ddof=1$ trap. 4. Print the TRUE population mean and standard deviation...

Write /report/03_step1.md (200-300 words)... Append a row to ANALYSIS_LOG.md.

Prompt 3 (Step 2 / CLT & CI):

Read PROJECT_BRIEF.md. Write scripts/04_step2_clt_ci.py. Use the skill `confidence-interval-builder`.

Part A — Sampling distribution/CLT demonstration: 500 samples of $n=40$... 1x3 panel $n=5,15,40$... normality test on the 500 sample means. Part B — Confidence interval: 95% t-CI from the single sample... VERIFICATION... coverage check across 500 resamples.

Write /report/04_step2.md (300-400 words)... Append a row to ANALYSIS_LOG.md.

Prompt 4 (Step 3 / Hypothesis Framing):

Read PROJECT_BRIEF.md and /report/02_eda_findings.md. Use the skill `hypothesis-framer`. Write /report/05_step3.md. No new data work is required here beyond citing the EDA evidence.

The business claim... "Our large-cap Indian IT basket has an average daily trading range of at most 2.0%..." Formalise this properly: (1) parameter and benchmark, (2) H_0/H_a , (3) tail, (4) alpha, (5) test choice, (6) assumptions, honestly assessed against independence.

Keep it under 600 words... Append a row to ANALYSIS_LOG.md.

Prompt 5 (Step 4 / Type I/II Error & Power):

Read PROJECT_BRIEF.md and /report/05_step3.md. Use the skill `type2-error-analyzer`. Write scripts/06_step4_errors.py and /report/06_step4.md.

Part A — business interpretation of Type I/Type II error in this context... which error the desk should fear. Part B — quantitative power analysis: statsmodels TTestPower across n and Cohen's d; observed effect size and power at n=40; min n for 80%/90% power; power curve figure; Monte Carlo verification.

Close with a paragraph on the sample-size/beta relationship... Append a row to ANALYSIS_LOG.md.

Prompt 6 (Step 5 / Critical Value vs p-Value):

Read PROJECT_BRIEF.md and /report/05_step3.md. Use the skill `hypothesis-test-decision-maker`. Write scripts/07_step5_decision.py and /report/07_step5.md.

Run the Step 3 hypothesis test... on a random sample of n=40 IT-sector rows... Approach 1 — critical value... Approach 2 — p-value... side-by-side decision table... Figure... Robustness check: Wilcoxon... VERIFICATION: cross-check the scipy t statistic and p-value...

Write /report/07_step5.md (250-350 words)... Append a row to ANALYSIS_LOG.md.

Prompt 7 (Step 6 / Three t-Tests):

Read PROJECT_BRIEF.md. Use the skill `two-population-t-tester`, extended for the paired case. Write scripts/08_step6_ttests.py and /report/08_step6.md. Use seed 42 for every draw and save every sample used to data/processed/.

TEST A — One-sample t-test... TEST B — Two-independent-sample t-test... Levene FIRST... pooled AND Welch... Mann-Whitney robustness. TEST C — Paired t-test... explain why genuinely paired... VERIFICATION: paired == one-sample on the difference column... Wilcoxon... Figure.

Write /report/08_step6.md (500-700 words)... Append a row to ANALYSIS_LOG.md.

Prompt 8 (Step 9 / Synthesis):

Read every file in /report/ and outputs/tables/08_ttest_results.csv. Do not run new analysis. Write /report/09_synthesis.md.

Produce ONE paragraph (180-250 words, no bullet points, no headings) that converts the Step 3 through Step 6 results into a single managerial recommendation... lead with the decision... cite specific numbers... state what changes operationally... state the confidence level and one honest caveat...

Then, separately, list the three most defensible sentences... and for each one name the exact table or figure that backs it. Append a row to ANALYSIS_LOG.md.

Prompt 9 (Step 10 / Sceptical Audit):

Read PROJECT_BRIEF.md. You are now acting as a sceptical reviewer auditing this project, not as its author. Assume errors are present and find them. Write scripts/10_audit.py and /report/10_audit.md.

1. Reproducibility: re-run every script... 2. Independent recomputation: ...using a DIFFERENT method... 3. Bootstrap cross-check... 4. Error hunt — ddof, tail direction, Welch vs pooled, paired alignment, sample sizes, terminology, figure/table labelling, unsupported claims. 5. List every place an AI-generated number was accepted without independent check...

Output a findings table... Then fix the blockers and majors, re-run the affected scripts, and show me the corrected values. Append a row to ANALYSIS_LOG.md.

Prompt 10 (Final Report Assembly — this deliverable):

Read every file in /report/ and every file in /outputs/. Assemble /report/FINAL_REPORT.md and /report/FINAL_REPORT.docx.

Hard constraint: the main body must fit in 8 pages... Structure of the main body: [ten numbered sections as above]... First page must include: [title, group, faculty, disclosure, skills table]... Appendix (unlimited): [A–G as below]... Quote Citations: APA 7th edition... Flag for me... anything you had to cut... and anything you are not fully confident is correct.

Appendix B — Definition of Every Skill Invoked

`sample-point-estimator` (from sample-point-estimator/SKILL.md): Parses the population array; selects a random sample of size n ; computes sample mean $\bar{x} = \sum x_i / n$, sample standard deviation $s = \sqrt{\sum (x_i - \bar{x})^2 / (n-1)}$, sample proportion (for binary categories, unused here); computes the standard error of the mean $s_{\bar{x}} = s / \sqrt{n}$. Used in Section 3 / Step 1.

`confidence-interval-builder` (from confidence-interval-builder/SKILL.md): Given n , mean, s (or σ), and confidence level: if σ is known, uses the z-distribution, interval = mean $\pm z_{(\alpha/2)} \cdot (\sigma / \sqrt{n})$; if σ is unknown, uses Student's t with $df = n-1$, interval = mean $\pm t_{(\alpha/2)} \cdot (s / \sqrt{n})$; outputs the margin of error and bounds. Used in Section 4 / Step 2.

`hypothesis-framer` (from hypothesis-framer/SKILL.md): Parses the business scenario; identifies the parameter of interest; determines the manager's claimed effect (H_a) and frames the status-quo baseline as H_0 ; classifies the test as one-tailed (upper or lower) or two-tailed. Used in Section 5 / Step 3.

`type2-error-analyzer` (from type2-error-analyzer/SKILL.md): Given μ_0 , μ_1 , σ , n , α , computes the critical mean under H_0 for an upper-tailed test ($x_{crit} = \mu_0 + z_{\alpha} \cdot \sigma / \sqrt{n}$), the z-value of that critical mean under the alternative, $\beta = P(Z < z)$, and power = $1 - \beta$. Used in Section 6 / Step 4 (extended in the script with statsmodels.stats.power.TTestPower for the t-distribution case and a Monte Carlo cross-check).

`hypothesis-test-decision-maker` (from hypothesis-test-decision-maker/SKILL.md): Given a test statistic, test type, and α , computes the p-value (upper-, lower-, or two-tailed) and applies both decision rules — reject H_0 if $p \leq \alpha$, or reject H_0 if the statistic exceeds the critical value — outputting the formal decision. Used in Section 7 / Step 5.

`two-population-t-tester` (from two-population-t-tester/SKILL.md): Given n_1 , n_2 , means, and standard deviations for two samples, computes the standard error (independent case: $\sqrt{s_1^2/n_1 + s_2^2/n_2}$, Welch–Satterthwaite df), the t-statistic, and compares against the critical value or p-value. Used in Section 8 / Step 6; extended in the script to also compute a paired t-test (as a one-sample t-test on the row-wise difference column) and pooled-variance and Mann-Whitney/Wilcoxon variants not in the original specification.

Appendix C — Raw Script Output for Every Step

Full, unedited console output from a fresh execution of each script (`00_verify_environment.py` and `02-08`, `10` were re-run at the time this report was assembled; `01_acquire_data.py`'s original acquisition ran before this session began (Step 0/1, prior agent) — see Appendix F for why it was not re-run wholesale for this appendix, and an excerpt from the verification re-run performed during the Step 10 audit is given instead).

00_verify_environment.py

```
=== Environment Verification ===
yfinance: 1.5.1
pandas: 3.0.3
numpy: 2.5.1
scipy: 1.18.0
statsmodels: 0.14.6
matplotlib: 3.10.9
```

01_acquire_data.py (excerpt from the Step 10 audit's verification re-run; full original output is not independently recoverable — see Appendix F, finding on data-source non-determinism)

```
[4] Performing India-Specific Data Hazards Checks...

* Hazard (a): Short Tickers / Fetch Failures:
Findings: All 12 tickers returned balanced row counts (Min: 494, Max: 494 rows per ticker).
Decision & Rationale: Kept (all 12 tickers fetched with full history)

* Hazard (a): LTIM.NS API Failure & Replacement:
Findings: LTIM.NS returned HTTP 404 (quote not found) from yfinance API. Re-fetched individually
and verified unavailable. Replaced with PERSISTENT.NS to maintain a balanced 6-ticker IT vs
6-ticker FMCG dataset (494 clean rows per ticker).
Decision & Rationale: Replaced LTIM.NS with PERSISTENT.NS

* Hazard (b): Corporate Actions / Ex-Dividend Discontinuities:
Findings: Identified 22 ex-dividend/bonus step dates across tickers where Close/Adj Close ratio
changed by >1%.
Decision & Rationale: Kept (unadjusted OHLC used for daily range and intraday move)

* Hazard (c): Zero/Stale Range_pct Sessions:
Findings: Identified 11 rows with range_pct <= 0.01% (stale quote/trading halt session on
2025-03-18 across 11 tickers).
Decision & Rationale: Kept & Reported (valid historical exchange records)

* Hazard (d): Special Low-Volume Sessions (Muhurat Trading):
Findings: Identified 3 low-volume dates (<30% median daily volume), including Diwali Muhurat
trading sessions (2024-11-01 and 2025-10-21).
Decision & Rationale: Kept & Reported (authentic exchange trading sessions)

--- DESCRIPTIVE STATISTICS FOR KEY VARIABLES ---
              count      mean      std      min      25%      50%      75%      max
range_pct      5928.0  2.070591  1.077912  0.000000  1.374976  1.828823  2.450564  14.580113
daily_return_pct  5928.0  0.025063  1.478353 -9.857074 -0.758125 -0.010134  0.764529  10.867159
abs_overnight_gap_pct  5928.0  0.447739  0.635429  0.000000  0.089719  0.290186  0.562924  10.000108
abs_intraday_move_pct  5928.0  0.990352  0.914657  0.000000  0.352619  0.752185  1.366886  10.233905

Wrote report/data_dictionary.md successfully.
```

02_eda.py

```
=====
                        STEP 02: EXPLORATORY DATA ANALYSIS (EDA)
=====

Loaded data/processed/population.csv: 5928 rows

[1] Computing summary statistics for range_pct (overall + by sector)...
```

```
-- Overall (n=5928) --
mean: library=2.0705908114 raw=2.0705908114 diff=0.00e+00
std: library=1.0779122556 raw=1.0779122556 diff=0.00e+00
skewness: library=2.4591177185 raw=2.4591177185 diff=4.44e-16
kurtosis: library=11.8530733690 raw=11.8530733690 diff=1.78e-15

-- Technology (n=2964) --
mean: library=2.1905503410 raw=2.1905503410 diff=0.00e+00
std: library=1.1342039425 raw=1.1342039425 diff=0.00e+00
skewness: library=2.5110533966 raw=2.5110533966 diff=8.88e-16
kurtosis: library=12.7738723054 raw=12.7738723054 diff=8.88e-15

-- Consumer Staples (n=2964) --
mean: library=1.9506312818 raw=1.9506312818 diff=0.00e+00
std: library=1.0044764419 raw=1.0044764419 diff=0.00e+00
skewness: library=2.3671924823 raw=2.3671924823 diff=1.33e-15
kurtosis: library=10.0018899121 raw=10.0018899121 diff=3.55e-15

Saved outputs/tables/02_summary_stats.csv

[3] Formal normality checks on range_pct...
Shapiro-Wilk (n=500 subsample): W = 0.782859, p = 3.012416e-25
D'Agostino K^2 (full n=5928): K2 = 3337.012856, p = 0.000000e+00
Anderson-Darling (full n=5928): A2 = 210.006210
significance level 15.0% critical value = 0.5610 -> REJECT normality
significance level 10.0% critical value = 0.6310 -> REJECT normality
significance level 5.0% critical value = 0.7520 -> REJECT normality
significance level 2.5% critical value = 0.8730 -> REJECT normality
significance level 1.0% critical value = 1.0350 -> REJECT normality

[4] Levene's test for equality of variances (Technology vs Consumer Staples)...
Technology variance: library=1.2864185832 raw=1.2864185832 diff=0.00e+00
Consumer Staples variance: library=1.0089729223 raw=1.0089729223 diff=0.00e+00
Levene's test (center=median): W = 15.583448, p = 7.985669e-05

Saved outputs/tables/02_normality_variance_tests.csv
=====
STEP 02 COMPLETE
=====
```

03_step1_sampling.py

```
=====
STEP 03: SIMPLE RANDOM SAMPLE & POINT ESTIMATES (skill: sample-point-estimator)
=====

Loaded population: 5928 rows

[1] Drawing simple random sample without replacement (n=40, seed=42)...
Saved data/processed/sample_n40.csv (40 rows)

[2] Computing point estimates (skill: sample-point-estimator)...
Sample mean (x-bar) = 2.004123
Sample std dev (s, ddof=1) = 1.159703
Standard error (s/sqrt(n)) = 0.183365
n = 40
Degrees of freedom (n-1) = 39

[3] VERIFICATION: recomputing from first principles (plain numpy, no pandas convenience)...

quantity          skill output    manual output    abs diff
sample_mean (x-bar)    2.004123    2.004123    0.00e+00
sample_std (s, ddof=1)  1.159703    1.159703    0.00e+00
standard_error (s/sqrt(n))  0.183365    0.183365    0.00e+00
n                    40.000000    40.000000    0.00e+00
degrees_of_freedom (n-1)  39.000000    39.000000    0.00e+00

--- ddof=0 vs ddof=1 trap ---
```

```

std with ddof=1 (sample, Bessel-corrected): 1.159703
std with ddof=0 (population formula):      1.145115
CORRECT CHOICE: ddof=1.

[4] TRUE population values (for context only)...
TRUE population mean (mu)      = 2.070591
TRUE population std dev       = 1.077912
Sampling error (x-bar - mu)    = -0.066468

Saved outputs/tables/03_point_estimates.csv
=====
                        STEP 03 COMPLETE
=====

```

04_step2_clt_ci.py

```

=====
STEP 04: CLT DEMONSTRATION & CONFIDENCE INTERVAL (skill: confidence-interval-builder)
=====

Loaded population: 5928 rows. True mu = 2.070591, true sigma = 1.077821

[A2] Comparing empirical vs theoretical standard error (n=40)...
Mean of 500 sample means      = 2.074436 (true mu = 2.070591)
Empirical SE (std of means)   = 0.171737
Theoretical SE (sigma/sqrt40) = 0.170419

[A5] Normality test on the 500 sample means (n=40 case)...
Shapiro-Wilk on 500 sample means: W = 0.989357, p = 1.085087e-03

[B6] Building 95% CI from the single original sample...
Sample mean = 2.004123, s = 1.159703, SE = 0.183365, df = 39
t critical value (alpha=0.05, df=39) = 2.022691
Margin of error = 0.370891
95% CI (skill) = [1.633232, 2.375014]

[B7] VERIFICATION: manual recomputation of the t-interval...


| quantity         | skill output | manual output | abs diff |
|------------------|--------------|---------------|----------|
| t_critical_value | 2.022691     | 2.022691      | 0.00e+00 |
| margin_of_error  | 0.370891     | 0.370891      | 0.00e+00 |
| ci_lower         | 1.633232     | 1.633232      | 0.00e+00 |
| ci_upper         | 2.375014     | 2.375014      | 0.00e+00 |



z-based interval (for contrast): z_crit=1.959964, MOE=0.359389, CI = [1.644734, 2.363512]

[B8] Coverage check: 95% CI computed for each of the 500 resamples (n=40)...
Empirical coverage (500 CIs) = 93.80% (469/500 intervals contained the true mean)

Saved outputs/tables/04_clt_ci_results.csv
=====
                        STEP 04 COMPLETE
=====

```

06_step4_errors.py

```

=====
STEP 06: TYPE II ERROR & POWER ANALYSIS (skill: type2-error-analyzer)
=====

Actual sample (n=40): mean = 2.004123, s = 1.159703

[5] Observed effect size and power of the actual n=40 test...
Observed Cohen's d = (2.004123 - 2.0) / 1.159703 = 0.003555
Power at n=40, d=0.003555, alpha=0.05: power = 0.052321
Beta (Type II error probability) = 1 - power = 0.947679

[6] Minimum n required for 80% and 90% power at the observed effect size...
n required for 80% power: 489,075.2

```



```

n required for 90% power: 677,448.2

[8] VERIFICATION: Monte Carlo power simulation for one grid cell (n=40, d=0.5)...
Analytic power (statsmodels TTestPower): 0.928106
Monte Carlo power (10,000 sims, seed=42): 0.933500 (9335/10000 rejected)
Absolute difference: 0.005394

Saved outputs/tables/06_observed_power_results.csv
=====
STEP 06 COMPLETE
=====

```

07_step5_decision.py

```

=====
STEP 07: HYPOTHESIS TEST DECISION (skill: hypothesis-test-decision-maker)
=====

Loaded population: 5928 rows (2964 Technology rows)

[Setup] Drawing SRS without replacement, n=40, seed=42, from Technology rows only...
Sample mean = 2.307385, s = 1.166197, SE = 0.184392, df = 39

[Approach 1] Critical value method...
Critical t value (upper tail): t_crit = 1.684875
Observed test statistic: t = 1.667022
Decision: t = 1.667022 < t_crit = 1.684875 -> FAIL TO REJECT H0
Critical value in original units: x_crit = 2.310677%

[Approach 2] p-value method...
One-tailed p-value = 0.051761
Decision: p = 0.051761 > alpha = 0.05 -> FAIL TO REJECT H0

[3] Robustness check: Wilcoxon signed-rank test...
W = 478.000000, p = 0.184100 -> FAIL TO REJECT H0 (unchanged)

[4] VERIFICATION: manual cross-check via scipy.stats.t.sf / .ppf only...
scipy.stats.ttest_1samp: t = 1.667022, p = 0.051761
Manual (explicit sum/(n-1) arithmetic + t.sf): t = 1.667022, p = 0.051761
Diff (t): 0.00e+00 Diff (p): 0.00e+00

Saved outputs/tables/07_hypothesis_test_results.csv
=====
STEP 07 COMPLETE
=====

```

08_step6_ttests.py

```

=====
STEP 08: THREE t-TESTS (skill: two-population-t-tester, extended)
=====

===== TEST A: ONE-SAMPLE t-TEST =====
n=40, mean=2.307385, s=1.166197, SE=0.184392, df=39
t=1.667022, t_crit=1.684875, p=0.051761 -> FAIL TO REJECT H0
95% CI for mean = [1.934417, 2.680353]
VERIFICATION: t statistic skill=1.667022 manual=1.667022 diff=0.00e+00

===== TEST B: TWO-INDEPENDENT-SAMPLE t-TEST =====
Levene's test for equal variances: W=0.188885, p=0.665046
FAIL TO REJECT equal variances -> pooled t-test adopted.
Pooled t-test: t=-0.781711, df=78, p=0.436751
Group means: Technology=2.083976 (s=1.298101), Consumer Staples=2.319746 (s=1.397716)
Cohen's d = -0.174796
Robustness -- Mann-Whitney U: U=694.000000, p=0.310023 -> FAIL TO REJECT H0 (unchanged)

===== TEST C: PAIRED t-TEST =====
Mean abs_overnight_gap_pct = 0.505348, Mean abs_intraday_move_pct = 0.947642

```

```

Mean difference = -0.442294, s(diff) = 0.899912, SE(diff) = 0.142289, df=39
Paired t-test: t=-3.108430, p=0.003504
One-sample t-test on diff column (equivalence): t=-3.108430, p=0.003504 [EXACT MATCH]
Decision: REJECT H0. Cohen's d (paired) = -0.491486
Robustness -- Wilcoxon signed-rank: W=179.000000, p=0.001441 -> REJECT H0 (unchanged)

```

```

===== CONSOLIDATED RESULTS TABLE =====
test                H0                Ha                n statistic df p_value
decision effect_size
A: One-sample t-test mu <= 2.0          mu > 2.0          40  1.667022  39 0.051761 FAIL TO
REJECT H0          0.263579
B: Two-sample t-test (pooled) mu_IT = mu_FMG mu_IT != mu_FMG 40+40 -0.781711  78 0.436751 FAIL TO
REJECT H0          -0.174796
C: Paired t-test    mu_diff = 0          mu_diff != 0      40  -3.108430  39 0.003504
REJECT H0          -0.491486
=====
STEP 08 COMPLETE
=====

```

10_audit.py

```

=====
STEP 10: SCEPTICAL AUDIT OF THE FULL PROJECT
=====

===== 1. REPRODUCIBILITY =====
[1a] adj_close differs by up to 0.00146 on 1,981/5,928 rows between live pulls; OHLC and all
used variables byte-identical; hazard(b) count unchanged (22=22).
[1b] Compared 23 output files across a fresh re-run of 6 scripts.
PASS: every table and sample file regenerated BYTE-IDENTICAL.

===== 2. INDEPENDENT RECOMPUTATION =====
               quantity original recomputed abs_diff
Step1 sample_mean 2.004123  2.004123 0.000000e+00
Step1 sample_std  1.159703  1.159703 0.000000e+00
Step1 SE          0.183365  0.183365 0.000000e+00
Step2 CI lower    1.633232  1.633232 0.000000e+00
Step2 CI upper    2.375014  2.375014 0.000000e+00
Step5 t_statistic 1.667022  1.667022 0.000000e+00
Step5 p_value     0.051761  0.051761 0.000000e+00
Step6-B t_statistic (pooled, current fixed sample) -0.781711 -0.781711 2.997602e-15
Step6-B p_value (pooled, current fixed sample) 0.436751 0.436751 1.609823e-15
Step6-C t_statistic -3.108430 -3.108430 4.440892e-16
Step6-C p_value     0.003504  0.003504 4.336809e-19
Max absolute difference across every independently-recomputed quantity: 3.00e-15

===== 3. BOOTSTRAP CI CROSS-CHECK =====
Bootstrap 95% percentile CI (10,000 resamples, seed=42): [1.677295, 2.389540]
t-based 95% CI (Step 2, same sample): [1.633232, 2.375014]
Sample skewness of range_pct in this n=40 draw: 1.9533

===== 4. ERROR HUNT =====
[4a] ddof=0 occurrences found: 4 (all legitimate; see population-sigma inconsistency finding)
[4b] Tail direction: PASS for all tests.
[4c] Welch vs pooled: Levene p=0.665046 -> expected pooled; script adopted: pooled. PASS
[4d] Paired alignment: PASS (same dataframe, positional).
[4e] Sample sizes: all PASS (n=40 confirmed for every saved sample file).
[4f] Terminology: PASS at report level.
[4g] Figure/table labelling: 2 figure gaps + 3 missing-n tables found (minor).
[4h] Unsupported claims: 2 blockers + 1 major found and fixed (see Appendix F).
[4i] Cross-test independence: MAJOR bug found and fixed (see Appendix F).

===== 5. UNCHECKED NUMBERS =====
Cohen's d (Test B, Test C): checked via t-d identity, diff=0.00e+00 both.
Step 4 observed-power: checked via Monte Carlo, diff=0.000521.

===== CORRECTED VALUES (Test B, before vs after fix)
=====

```

quantity	before_fix	after_fix
Levene p-value	0.027874	0.665046
Adopted test	Welch	pooled
t statistic	1.963716	-0.781711
p-value	0.053816	0.436751
Cohen's d	0.4391	-0.174796
Mean diff (IT-FMCG)	0.426926	-0.23577
=====		
STEP 10 COMPLETE		
=====		

Appendix D — Data Dictionary and Download Manifest

Data Dictionary (report/data_dictionary.md)

Column	Definition	Units	Formula	Source
date	Trading session date	YYYY-MM-DD	Yahoo Finance index	Yahoo Finance (NSE)
ticker	NSE symbol	Categorical	.NS suffix	NSE India
sector	Sector classification	Categorical	Technology / Consumer Staples	Project classification
open, high, low, close	Raw unadjusted OHLC prices	INR (₹)	—	Yahoo Finance (NSE)
adj_close	Adjusted close (dividends/splits)	INR (₹)	—	Yahoo Finance (NSE)
volume	Shares traded	Count	—	Yahoo Finance (NSE)
range_pct	Relative daily trading range	%	$(\text{high} - \text{low}) / \text{open} \times 100$	Derived (primary variable)
daily_return_pct	Close-to-close return	%	$(\text{close}_t - \text{close}_{t-1}) / \text{close}_{t-1} \times 100$	Derived
overnight_gap_pct	Overnight price gap	%	$(\text{open}_t - \text{close}_{t-1}) / \text{close}_{t-1} \times 100$	Derived
intraday_move_pct	Intraday price change	%	$(\text{close}_t - \text{open}_t) / \text{open}_t \times 100$	Derived
abs_overnight_gap_pct	\	overnight_gap_pct\		%
abs_intraday_move_pct	\	intraday_move_pct\		%
year, month, weekday	Calendar parts of date	—	—	Derived

Download Manifest (data/raw/DOWNLOAD_MANIFEST.txt)

DOWNLOAD MANIFEST

Download Timestamp (UTC): 2026-07-22 17:00:03 UTC
yfinance Version: 1.5.1
Requested Tickers (12): TCS.NS, INFY.NS, HCLTECH.NS, WIPRO.NS, TECHM.NS, PERSISTENT.NS, HINDUNILVR.NS, ITC.NS, NESTLEIND.NS, BRITANNIA.NS, DABUR.NS, MARICO.NS
Requested Date Range: 2024-01-01 to 2026-01-01
Total Raw Rows Downloaded: 5940
Failed Tickers: None
Raw Data Cols: ['Adj Close', 'Close', 'High', 'Low', 'Open', 'Volume', 'Ticker', 'Date']

Rows per ticker (all 494, balanced): six IT tickers (HCLTECH.NS, INFY.NS, PERSISTENT.NS, TCS.NS, TECHM.NS, WIPRO.NS) and six FMCG tickers (BRITANNIA.NS, DABUR.NS, HINDUNILVR.NS, ITC.NS, MARICO.NS, NESTLEIND.NS), each spanning 2024-01-02 to 2025-12-31.

Appendix E — ANALYSIS_LOG.md, In Full

Time stamp	Project step	Prompt summary	Tool/skill used	What was verified manually	Outcome
2026-	Step 0:	Set up directory	write_to_file,	Directory tree created,	Verified yfinance

07-22 22:20: 15	Workspace Setup	structure, constants, rules, analysis log, install dependencies	run_command	requirements.txt written, package versions verified	1.5.1, pandas 3.0.3, numpy 2.5.1, scipy 1.18.0, statsmodels 0.14.6, matplotlib 3.10.9
2026-07-22 22:30:00	Step 1: Data Acquisition & Processing	Download OHLCV for 12 NSE tickers, engineer variables, hazard checks, save population.csv	yfinance, pandas, numpy, scripts/01_acquire_data.py	5,928 clean rows (494/ticker), 12 rows dropped, zero formula diffs (>1e-9 check)	Built data/raw/yfinance_raw.csv, data/processed/population.csv, outputs/tables/*.csv, report/data_dictionary.md
2026-07-23 09:15:00	Step 2: EDA	Distributional profile of range_pct, normality tests, Levene's test	scipy.stats, scripts/02_eda.py	Dual-computed mean/std/skewness/kurtosis, all diffs < 1e-14	All normality tests reject normality; Levene rejects equal variances (p=8.0e-05); report/02_eda_findings.md (329 words)
2026-07-23 09:45:00	Step 3: Sampling & Point Estimates	SRS n=40 seed=42, point estimates via sample-point-estimator, ddof trap	scripts/03_step1_sampling.py	All 5 quantities matched manual recomputation, 0.00e+00 diff	$\bar{x}=2.0041$, $s=1.1597$, SE=0.1834; report/03_step1.md (293 words)
2026-07-23 10:20:00	Step 4: CLT & CI	500 resamples, CLT panel, 95% t-CI via confidence-interval-builder, coverage check	scripts/04_step2_clt_ci.py	Manual t-critical/MOE/CI recompute, 0.00e+00 diff	95% CI=[1.633,2.375]; coverage=93.8%; report/04_step2.md (392 words)
2026-07-23 10:50:00	Step 5: Hypothesis Framing	Formalise "IT range <= 2.0%" via hypothesis-framer — report-only	—	Cross-checked narrative claims against existing EDA evidence	H0/Ha, one-tailed test, alpha=0.05; report/05_step3.md (595 words)
2026-07-23 11:20:00	Step 6: Type I/II Error & Power	Power grid, observed power/beta, Monte Carlo, via type2-error-analyzer	statsmodels.stats.power.TTestPower	MC power (10k sims) vs analytic, diff 0.0054	Observed $d \approx 0.0036$, power=0.0523; report/06_step4.md
2026-07-23 11:50:00	Step 7: Hypothesis Test Decision	Critical value + p-value on fresh IT sample via hypothesis-test-decision-maker	scipy.stats	Manual cross-check of t/p, 0.00e+00 diff	p=0.0518, fail to reject; report/07_step5.md (326 words)
2026-07-23 12:30:00	Step 8: Three t-Tests	One-sample, two-sample (Levene→pooled/Welch), paired, via two-population-t-tester	scipy.stats	Recomputed all 3 t-statistics by hand, diffs 0.00e+00–2.22e-16	Test C rejects H0; report/08_step6.md (641 words, later revised)
2026-	Step 9:	One-paragraph	—	Cross-checked every	report/09_synthesis.

07-23 13:00:00	Managerial Synthesis	recommendation, report-only		cited number against outputs/tables/08_ttest_results.csv	md (243 words, later revised)
2026-07-23 14:00:00	Step 10: Sceptical Audit	Reproducibility, independent recomputation, bootstrap, error hunt	scripts/10_audit.py	Full pipeline re-run, statsmodels cross-check, MC verification	Found & fixed 2 blockers + 2 majors; flagged 5 minors; report/10_audit.md

Appendix F — Step 10 Audit Findings and Resolutions

Issue	Severity	File(s)	Resolution
scripts/07_step5_decision.py saved no output table — every number in report/07_step5.md existed only as console output	Blocker	scripts/07_step5_decision.py	Fixed: added outputs/tables/07_hypothesis_test_results.csv
Step 4's observed effect size, power, beta, min-n, and Monte Carlo check were printed but never saved	Blocker	scripts/06_step4_errors.py	Fixed: added outputs/tables/06_observed_power_results.csv
Test B's Levene stat/p and raw group means/stds weren't captured in any saved table	Major	scripts/08_step6_ttests.py	Fixed: added outputs/tables/08_testB_group_stats.csv
Test A's sample and Test B's original IT draw were byte-identical (confirmed by file diff) — both scripts drew the IT-filtered sample first from a fresh rng(42), so Test A and Test B's IT arm were silently the same 40 rows, not independent evidence	Major	scripts/08_step6_ttests.py, report/08_step6.md	Fixed: reordered Test B (FMCG drawn first, then IT); re-ran; updated report/08_step6.md and report/09_synthesis.md with the corrected, genuinely independent result (which reverses direction and loses significance)
CLT panel figure sets the y-axis label on only 1 of 3 subplots	Minor	scripts/04_step2_clt_ci.py	Not fixed (documented only, per audit scope)
Q-Q plot uses scipy's default axis labels, missing the "(%)" unit	Minor	scripts/02_eda.py	Not fixed (documented only)
"True population std" computed with ddof=1 in one script (1.077912) and ddof=0 in another (1.077821) for the same quantity	Minor	scripts/03_step1_sampling.py, scripts/04_step2_clt_ci.py	Not fixed (documented only)
Live data source is not byte-reproducible on re-run (adj_close drifts; OHLC and all used variables unaffected)	Minor	scripts/01_acquire_data.py	Not fixed (documented; canonical committed data treated as source of truth)
Three output tables report statistics without an explicit n column	Minor	scripts/02_eda.py, scripts/04_step2_clt_ci.py	Not fixed (documented only)

Appendix G — Package Versions and Random Seed

RANDOM_SEED = 42, fixed at the top of every script.

Package	Version
yfinance	1.5.1
pandas	3.0.3

numpy	2.5.1
scipy	1.18.0
statsmodels	0.14.6
matplotlib	3.10.9